

# The Build-Out Map

US & Nasdaq companies wired into the machine you described — chips, minerals, power, robots, and orbit.

You laid out a single connected thesis: Bitcoin pushing toward \$250k by end-2032, an explosion of AI companies and data centers, rare high-cost minerals feeding robotics and conductors, the construction firms that pour the concrete and wire the power, the fuel of starships and satellites, and the companies that build the actual *joints of robots*. This dossier maps that whole chain to real, currently-trading US companies — the giants to believe in and hold, plus the newborns with true potential. Every number here is sourced from May 2026 filings and market data.

**The one sentence that connects everything:** The same force — exponential demand for compute — is what lifts memory chips, drains the power grid (forcing new nuclear + construction), pulls rare minerals out of the ground for magnets and chips, and funds the robots and satellites on top. Even Bitcoin miners have become AI landlords. You are not picking 6 themes; you are picking 6 floors of *one building*.

## 1 Semiconductors & Memory LIKE SANDISK — YOUR ANCHOR

This is the floor you already feel. AI doesn't just need GPUs — it needs vast oceans of **memory** next to them (HBM, DRAM, NAND). Supply is sold out into 2027, and the shortage may run to 2030. TrendForce sees memory revenue jumping **134% to \$552B in 2026**.

| COMPANY        | TICKER            | WHAT IT IS                                             | THE FACT                                                                                   |
|----------------|-------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------|
| SanDisk        | <b>SNDK</b>       | NAND flash storage (spun out of Western Digital)       | +552% YTD in 2026; revenue +251% YoY to \$5.95B last quarter                               |
| Micron         | <b>MU</b>         | The only major US-listed pure memory maker; HBM for AI | +179% YTD; HBM capacity booked into 2027; record Q2 FY26                                   |
| ASE Technology | <b>ASX</b>        | The chip packaging/test house you found & love         | World's largest outsourced assembly & test — every advanced chip passes through this layer |
| Nvidia / AMD   | <b>NVDA / AMD</b> | The AI accelerators memory is built around             | The demand engine that makes every row above necessary                                     |

**Why hold:** Memory is the bottleneck that is hardest to fake or rush — you can't spin up a fab in a year. SanDisk and Micron are the purest US listings on a shortage that the industry itself says lasts to 2030. ASX is the "toll booth" — it doesn't matter whose chip wins, it gets packaged through that layer.

## 2 Critical Minerals & Rare Earths BETTING ON THE USA

Your instinct is exactly right: robots need magnets, magnets need rare earths (Nd, Dy, Tb), and chips need gallium, germanium, antimony, high-purity quartz. China controls most of it — so the US is now *price-flooring and funding* a domestic chain. In 2026 the government launched "Project Vault": **\$1.67B private capital + a \$10B Export-Import Bank loan** to stockpile these materials. This is policy tailwind, not hope.

| COMPANY | TICKER | WHAT IT IS | THE FACT / ARTEFACT |
|---------|--------|------------|---------------------|
|---------|--------|------------|---------------------|

|                    |             |                                           |                                                                                              |
|--------------------|-------------|-------------------------------------------|----------------------------------------------------------------------------------------------|
| MP Materials       | <b>MP</b>   | The flagship US rare-earth mine-to-magnet | Named repeatedly as <i>the</i> rare-earth bottleneck play for robotics                       |
| USA Rare Earth     | <b>USAR</b> | Mine-to-magnet, Round Top (Texas)         | Commercial production targeted 2028; picked Fluor + WSP as EPC in Jan 2026                   |
| Perpetua Resources | <b>PPTA</b> | Antimony + gold, Stibnite (Idaho)         | Secured a <b>\$2.9B US gov loan</b> (May 2026) — can supply ~35% of US antimony demand       |
| Ramaco Resources   | <b>METC</b> | Pivoting coal → rare earths + gallium     | Future revenue aimed at high-purity gallium, alumina & quartz — straight into semiconductors |

#### NEWBORN WITH TRUE POTENTIAL

**USA Rare Earth (USAR)** and **Serra Verde (SVRE)** are the young ones. Serra Verde is the *only* large-scale producer outside Asia of all four magnetic rare earths including the heavy ones (Dy, Tb, Y) that go into robot and drone motors — production started 2024 and is scaling now. These are early-stage; the artefact to watch is the offtake contract — when a magnet maker or automaker signs to buy their output, that's your fact.

**Why hold:** A robot can't move without a magnet, and a magnet can't be built without these metals. If "betting on the USA" is your edge, this is the most literal expression of it — Washington is now writing checks to make sure these mines exist.

## 3 Data-Center Construction & Power THE CONCRETE & THE GRID

Every AI dollar eventually becomes a building that needs power and cooling. Wall Street estimates AI infrastructure capex passes **\$1 trillion by 2028** and a cumulative **\$5–7 trillion by 2030**. The companies that *pour, wire, cool, and power* these don't need to pick the winning AI model — they get paid either way.

### The builders & the "plumbing"

| COMPANY                 | TICKER      | ROLE                                                      | THE FACT                                                               |
|-------------------------|-------------|-----------------------------------------------------------|------------------------------------------------------------------------|
| Vertiv                  | <b>VRT</b>  | Power, cooling & IT infrastructure inside the data center | Bought ThermoKey (Mar 2026) to expand AI-data-center heat rejection    |
| Comfort Systems USA     | <b>FIX</b>  | Mechanical/HVAC contractor for data centers               | Top-named non-tech beneficiary of the 2026 build-out                   |
| Sterling Infrastructure | <b>STRL</b> | Site prep, foundations, electrical groundwork             | Q1'26 revenue \$825M vs \$603M expected; raised FY guide to \$3.7–3.8B |
| EMCOR Group             | <b>EME</b>  | Electrical & mechanical construction                      | Record backlog                                                         |

### The power — nuclear & SMRs

AI is draining the grid, so the most under-owned piece is generation. SMR offtake deals tied to data centers jumped from 25 GW (end-2024) to **45 GW by April 2026**.

| COMPANY              | TICKER      | ROLE                                   | THE FACT                                                            |
|----------------------|-------------|----------------------------------------|---------------------------------------------------------------------|
| Constellation Energy | <b>CEG</b>  | Largest US nuclear fleet (21 reactors) | \$25.5B 2025 revenue; signing long-term data-center power contracts |
| Vistra               | <b>VST</b>  | Power generator for data-center load   | ~24x P/E; positioned for the demand surge                           |
| Oklo                 | <b>OKLO</b> |                                        |                                                                     |

|                |            |                                           |                                                                        |
|----------------|------------|-------------------------------------------|------------------------------------------------------------------------|
|                |            | Micro-reactors purpose-built for AI sites | Won US gov contract to build uranium fuel supply lines to data centers |
| NuScale Power  | <b>SMR</b> | Only NRC-certified SMR design in the US   | The reference name in modular nuclear                                  |
| Uranium Energy | <b>UEC</b> | Domestic uranium fuel supply              | The "shovel" behind every reactor above                                |

#### NEWBORN WITH TRUE POTENTIAL

**Oklo (OKLO)** and **Applied Digital (APLD)** are the young, high-belief plays — Oklo on micro-reactors for AI campuses, APLD as a pure AI/HPC data-center landlord. Pre-profit and volatile, but if the \$7T build-out is real, these are levered to it.

## 4 Robots — the Joints & Actuators **THE HARDEST PART TO FIND IN THE USA**

You asked specifically for companies that make the **joints of robots** — and here is the honest, important finding: the *actuator/joint* supply chain is today dominated by Asia and Germany (Harmonic Drive Systems – Japan; Nabtesco – Japan; Schaeffler – Germany; Sanhua – China, which just took a \$685M Tesla Optimus order). Pure US-listed joint makers barely exist yet. So the US-traded ways to own this theme are: (a) the brain & integrator, and (b) the raw materials feeding the joint.

| COMPANY      | TICKER      | ROLE IN A ROBOT                                | THE FACT                                                   |
|--------------|-------------|------------------------------------------------|------------------------------------------------------------|
| Tesla        | <b>TSLA</b> | Optimus — the integrator; final assembly in US | Optimus V3 prototype targeted Q1 2026; goal of 1M units/yr |
| Ambarella    | <b>AMBA</b> | Vision chips — the robot's eyes                | Named a core component bottleneck                          |
| Qualcomm     | <b>QCOM</b> | Edge-AI compute — the robot's reflexes         | Core component play for on-device intelligence             |
| MP Materials | <b>MP</b>   | The magnets inside every joint motor           | Listed alongside actuators as a production bottleneck      |

#### HOW TO TRACK A US "JOINT" NEWBORN

The KOID humanoid-robotics ETF is currently the cleanest single way to own the global joint makers (Harmonic Drive, Nabtesco, Schaeffler) from a US listing — its actuator holdings are up +89% to +135% since inception. **The artefact to hunt for:** a US-listed company announcing it has been designed into Tesla Optimus, Figure, or a Boston-Dynamics-class robot's actuator stack. The day a small US firm wins that socket is the day this becomes your SanDisk-style hold. Until then, own the magnet (MP) and the eyes/brain (AMBA, QCOM).

**Straight talk (you asked for facts):** there is no US-listed pure "robot joint" company to fall in love with yet. That's not a gap in the research — it's the actual state of the market in May 2026, and it's exactly why the first US name to break into the Optimus actuator chain could be enormous.

## 5 Starship Fuel, Satellites & Space **ORBIT**

The SpaceX IPO chatter (a rumored ~\$2T listing) is pulling the whole sector up. The public, investable names cluster around launch, landers, propulsion, and satellite broadband.

| COMPANY    | TICKER      | WHAT IT IS | THE FACT |
|------------|-------------|------------|----------|
| Rocket Lab | <b>RKLB</b> |            |          |

|                      |             |                                                 |                                                                                          |
|----------------------|-------------|-------------------------------------------------|------------------------------------------------------------------------------------------|
|                      |             | Launch + satellite propulsion (Neutron rocket)  | Q1'26 record revenue \$200.3M (+63.5%); \$2.2B backlog; launched Gauss electric thruster |
| Intuitive Machines   | <b>LUNR</b> | Lunar landers & surface delivery (NASA Artemis) | \$180.4M NASA contract; \$943M backlog; FY26 revenue guide \$900M–1B                     |
| AST SpaceMobile      | <b>ASTS</b> | Space-based cellular broadband                  | FCC nod for 248 satellites; \$1.2B+ contracted revenue                                   |
| Voyager Technologies | <b>VOYG</b> | Propellant & rocket-motor control tech          | Won DARPA "Burn n' Go" Phase 2 (\$16.5M) for tailorable solid-rocket thrust              |

#### CLOSEST MATCH TO YOUR "FUEL OF STARSHIPS"

**Voyager Technologies (VOYG)** is the literal propellant/propulsion play — its DARPA contract is for embedding controllable thrust into solid rocket motors. **Rocket Lab (RKLB)** is the more proven hold: it's now selling propulsion (Gauss thruster) *and* launch, the rare space company with real, growing revenue.

## 6 The Bitcoin → AI Bridge **YOUR \$250K THESIS, MADE INVESTABLE**

Here's the convergence that ties your crypto view to everything above: Bitcoin miners already own power contracts, land, and high-density facilities — so in 2026 they are **becoming AI data-center landlords**. Over **\$70B in cumulative AI/HPC contracts** have been signed across public miners; AI/HPC may be 70% of their revenue by year-end. If your Bitcoin-to-\$250k thesis is right, you win on the coin *and* the compute.

| COMPANY         | TICKER      | THE PIVOT                       | THE FACT                                                      |
|-----------------|-------------|---------------------------------|---------------------------------------------------------------|
| IREN            | <b>IREN</b> | Frontrunner BTC → AI conversion | 5-yr Microsoft deal ~\$1.94B/yr at ~85% project EBITDA margin |
| TeraWulf        | <b>WULF</b> | HPC hosting                     | \$12.8B contracted HPC revenue                                |
| Core Scientific | <b>CORZ</b> | CoreWeave compute landlord      | \$10.2B / 12-yr CoreWeave deal                                |
| Cipher Mining   | <b>CIFR</b> | AI hosting                      | Multi-billion deal with Google-backed Fluidstack              |

**One fact to respect:** these pivots are debt-funded (IREN ~\$3.7B convertibles; TeraWulf ~\$5.7B total debt). The contracts are real and huge — the leverage is the variable. That's the number to watch each quarter.

## The Shortlist — if you want belief-and-hold names

Distilling everything to the names with real revenue today (not just a story), mapped to your themes:

- Chips/memory (your SanDisk lane): **SNDK** , **MU** , **ASX**
- US minerals (betting on USA): **MP** (proven), **PPTA** (gov-funded), **USAR** (newborn)
- Build & power the data centers: **VRT** , **STRL** , **CEG** , **OKLO** (newborn)
- Robots (own the magnet + brain until a US joint maker emerges): **TSLA** , **AMBA** , **MP**
- Space & fuel: **RKLB** (proven), **VOYG** (propellant), **LUNR**
- Bitcoin → AI bridge: **IREN** , **CORZ** , **WULF**

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This dossier is factual research compiled from public May-2026 filings and market reporting to help you do your own deep dive — it is not a recommendation to buy or sell, and I'm not a financial advisor. Prices and contracts move fast; verify each fact against the latest filing before you act. Sources are listed in the chat message accompanying this file.